



SENIOR THESIS IN MATHEMATICS

Inference Without Data

A Causal Transportability Framework for Ketamine
Effects in Transgender and Gender-Diverse
Populations

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Submitted to Pomona College in Partial Fulfillment
of the Degree of Bachelor of Arts

April 30, 2026

Acknowledgments

I am deeply grateful to Professor Hardin for her guidance throughout this process, which strengthened both my communication and critical thinking skills. Thank you to my friends and family for their genuine interest in my work and for being an unwavering support system as I navigated so much uncertainty.

I want to acknowledge transgender people navigating the healthcare system, who face adversity every day as their identities are politicized and invalidated. This paper seeks to uplift that community.

Technology Statement

All statistical computation and simulation were performed in R (version 4.4.0; R Core Team, 2024). Data wrangling and visualization relied on the **tidyverse** suite of packages (Wickham et al., 2019). Multivariate normal sampling used the **MASS** package (Venables and Ripley, 2002). No proprietary statistical software was used.

This thesis was typeset in L^AT_EX using Overleaf. Claude (Anthropic), a large language model, was used to assist with debugging and developing R code. All statistical methodology, interpretation, and written content are the author's own.

Abstract

This thesis develops a causal data fusion framework for estimating ketamine’s effects on chronic pain and depression in transgender and gender-diverse (TGD) patients, a population absent from existing ketamine trials. The available evidence is split across two incompatible sources: a ketamine cohort in a predominantly cisgender sample, and an epidemiologic study characterizing baseline symptom burden in TGD adults with no ketamine exposure.

Using Pearl’s transportability theory, I formalize the conditions under which these sources can be combined. A selection diagram and S -admissibility analysis establish that the ketamine response mechanism is invariant across populations conditional on shared baseline covariates, yielding an identifiable transport formula for the target interventional distribution.

Additionally, a statistical account of pain and depression as coupled Bayesian inference processes motivates joint modeling of both outcomes and explains why baseline severity carries substantive predictive weight rather than functioning as a nuisance covariate.

Because neither study reports individual-level data, I derive an anchor constraint tying the response surface intercept to published source means and characterize remaining uncertainty via a constrained prior predictive Monte Carlo procedure. Sensitivity analyses reveal that the available data cannot support a transport adjustment: the transported estimate is nearly identical to the source mean response across the full range of prior assumptions tested.

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Chapter 1

Introduction

1.1 Motivation and Clinical Context

Transgender and gender-diverse (TGD) adults experience chronic pain and depression at rates substantially higher than the general population. Epidemiologic evidence documents elevated prevalence of both conditions in TGD communities, compounded by minority stress, barriers to affirming care, and high rates of trauma exposure (Tabernacki et al., 2025).

Chronic pain and depression are not independent in this population: they co-occur at rates far above chance, reinforce one another through shared neurobiological pathways, and jointly impair quality of life in ways that neither condition alone predicts.

Ketamine has emerged over the past decade as a clinically significant treatment for both chronic pain and treatment-resistant depression. Originally developed as a dissociative anesthetic, subanesthetic ketamine infusions have demonstrated rapid and sustained reductions in pain and depressive symptoms across a range of chronic pain etiologies (Pickering et al., 2025). Its mechanism differs from conventional analgesics and antidepressants, acting primarily through NMDA receptor antagonism and downstream modulation of glutamatergic signaling, making it a candidate treatment for patients who have not responded to standard-of-care options.

Despite this promise, TGD patients are effectively absent from existing ketamine trials. The clinical literature on ketamine for chronic pain and depression has been developed almost exclusively in cisgender samples, with gender identity rarely recorded and TGD individuals too few in any single

study to support subgroup analysis. TGD patients are underrepresented in clinical research broadly, a consequence of both historical exclusion and the practical difficulties of recruiting and retaining a population that has good reasons to distrust medical institutions.

The consequence is an epistemic problem for clinicians. A provider treating a TGD patient with refractory chronic pain and comorbid depression has no evidence base from which to project what ketamine is likely to do. They can extrapolate from a literature built on a population that differs from their patient in ways that may or may not matter for treatment response. This thesis is a methodological argument that principled extrapolation is possible, that the structure of the available evidence is sufficient to support a causal estimate of ketamine’s effects in TGD patients, and that making the assumptions behind that extrapolation explicit is preferable to either ignoring the gap or declining to engage with it.

1.2 The Data Fusion Problem

The available evidence on ketamine and TGD populations is split across two studies that are individually insufficient but jointly informative. Understanding why neither is enough, and why together they are, is the central conceptual problem of the thesis.

The first source is a large prospective multicenter observational ketamine cohort (Pickering et al., 2025) reporting 12-month outcomes for $n = 585$ chronic pain patients treated with subanesthetic ketamine infusions. The cohort provides detailed information about how pain and depression change under ketamine: baseline severity scores, monthly trajectories, and summary statistics stratified by sex and pain etiology. What it does not provide is any information about TGD patients. The cohort is drawn from specialized chronic pain clinics and is predominantly cisgender. It characterizes the ketamine response mechanism in a population that does not represent the target of inference.

The second source is an epidemiologic study of chronic pain and psychiatric comorbidity in TGD adults (Tabernacki et al., 2025). This study characterizes the prevalence, severity, and correlates of chronic pain and depression in the population we are interested in. It provides the distributional information needed to describe who TGD patients are at baseline. What it does not provide is any ketamine exposure. No treatment is administered, no

outcomes under intervention are observed, and the study cannot say anything directly about what ketamine would do in this population.

The ketamine cohort identifies the treatment response mechanism but in the wrong population. The TGD study characterizes the right population but contains no treatment. Combining them naively, by appending datasets or averaging effect sizes, would be unjustified: the populations differ in their covariate distributions, and there is no guarantee that a response estimated in one population applies to the other.

What is needed is a framework that can formally license the combination, specifying exactly what must be true of the two populations for the ketamine response identified in the source cohort to be transportable to TGD patients. Pearl’s transportability theory provides that framework (Pearl, 2009). The causal question, what would happen if TGD patients received ketamine, can be expressed as an interventional distribution in a target population where the intervention was never observed. Under conditions that can be stated precisely using selection diagrams and do-calculus, that distribution is identifiable from the two available sources. The remainder of the thesis derives those conditions, implements the resulting estimator, and characterizes the uncertainty that survives after all published constraints are applied.

1.3 Contributions and Roadmap

This thesis develops a causal framework for estimating ketamine’s effects on chronic pain and depression in TGD adults, a population absent from the clinical trials that generated the existing evidence base. The framework is built across four substantive chapters.

Chapter 2 develops a statistical account of pain perception as Bayesian inference over hidden bodily states. This framing, combined with the clinical literature on pain-depression comorbidity, motivates joint modeling of both outcomes and explains why baseline symptom severity is a clinically meaningful predictor of treatment response rather than a nuisance covariate.

Chapter 3 establishes the causal framework that makes data fusion possible. Using Pearl’s transportability theory, selection diagrams, and do-calculus, it derives the conditions under which the ketamine response mechanism identified in a cisgender clinical cohort can be transported to a TGD target population. The central result is an identifiable transport formula expressing the target interventional distribution as a mixture of source-population

response surfaces weighted by the TGD covariate distribution.

Chapter 4 translates the transport formula into a computable quantity. It introduces the generative model, derives the anchor constraint that ties the intercept to published summary statistics, and implements a constrained prior predictive Monte Carlo procedure to characterize the distribution of transported mean responses consistent with available evidence. Sensitivity analyses over unidentified structural parameters confirm that the dominant uncertainty lies in the source response surface, not in the target population assumptions. The main finding is that the transported estimate is statistically indistinguishable from the source mean. The TGD population differs from the source cohort in the severity of the baseline symptoms, but the uncertainty about how baseline severity predicts the response to treatment is much greater than the adjustment that difference would produce.

Chapter 5 situates this finding within broader questions of underrepresentation in clinical research, reemphasizes what the framework does and does not contribute when its central estimate coincides with naive extrapolation, and outlines the data collection that would sharpen the transported estimate.

Chapter 2

Pain and Depression as Statistical Objects

Chronic pain can be thought of as an inference as well as a physical sensation. Under this framing, the brain does not directly observe whether the body is damaged or threatened; it maintains beliefs about hidden bodily states and updates them against incoming sensory evidence. When the brain’s prior on bodily threat becomes too precise to be revised by new sensory evidence, pain may persist even after tissue has healed.

This chapter develops that claim formally, following the model of Eckert et al. (2022). The goal here is to establish why pain has the statistical structure it does, why baseline symptom severity is a clinically meaningful predictor of treatment response rather than merely a covariate to adjust for, and why joint modeling of pain and depression is justified. That predictor is the covariate vector $z = (z_{\text{pain}}, z_{\text{dep}})$ that anchors the transport formula in Chapter 3 and enters the response surface $E[\Delta \mid z, S = 0]$ in Chapter 4.

First, pain perception is formalized as Bayesian inference over hidden states, using a Hidden Markov Model in which the brain’s transition prior encodes expectations about persistence or recovery. Chronic pain arises when this prior becomes too narrow. Second, Dirichlet priors on transition and emission probabilities formalize how prior precision accumulates through experience, explaining why patients with longer symptom histories are harder to recalibrate and why baseline severity is a substantive predictor of treatment response.

Depression does not factor into this model, but later in the clinical literature on comorbidity. Pain and depression co-occur at rates far above chance

(Aaron et al., 2025), respond to overlapping treatments, and show correlated baseline severity. That covariance is the reason we model them as a bivariate outcome in Chapter 4, and why baseline depression belongs in the covariate vector Z alongside baseline pain. The HMM here gives a mechanistic account of the pain side. The joint structure follows from comorbidity, not from extending the HMM to cover both.

2.1 The Hidden Markov Model of Pain Perception

Following Eckert et al. (2022), pain perception is formalized as a Hidden Markov Model in which the brain infers a latent bodily state from noisy sensory evidence. The hidden state

$$H_t \in \{\text{pain, no pain}\}$$

represents the body’s true internal condition at time t : whether nociceptive systems are signaling actual threat. This state is never directly observed. The observable variable

$$V_t \in \{\text{noxious, harmless}\}$$

represents the sensory signal reaching the brain at time t . The brain’s task is to infer H_t from $V_{1:t}$.

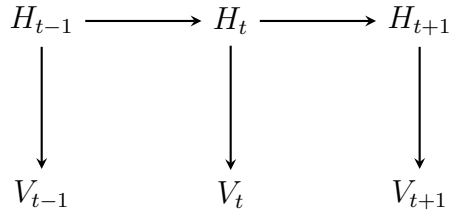


Figure 2.1: HMM graphical model for pain perception (Eckert et al., 2022). Hidden states H_t evolve over time and generate observable sensory signals V_t .

The graphical structure implies two conditional independence assumptions: each hidden state H_t depends only on H_{t-1} (the Markov property),

and each observation V_t depends only on H_t (emission independence). Together these yield the factorization of the joint distribution over hidden states and observations:

$$p(H_{1:T}, V_{1:T}) = p(H_1) \prod_{t=2}^T p(H_t | H_{t-1}) \prod_{t=1}^T p(V_t | H_t).$$

The three components are the prior over the initial pain state $p(H_1)$, the transition model $p(H_t | H_{t-1})$ encoding how likely pain is to persist or resolve, and the emission model $p(V_t | H_t)$ encoding how internal states generate sensory signals. The brain’s task is to invert this generative model: given observations $V_{1:T}$, infer the posterior over hidden states $p(H_{1:T} | V_{1:T})$. This inversion proceeds via the forward–backward algorithm, which computes the posterior recursively using the factorization above. The derivation is standard and omitted here; the chapter develops the transition prior further because that is the component that connects most directly to baseline severity and to the response surface in Chapter 4.

Transition and Emission Models

The factorization implies two sets of parameters. The transition model π_h encodes the brain’s prior belief about whether pain persists or resolves:

$$\pi_h = (p(H_t = \text{pain} | H_{t-1} = h), p(H_t = \text{no pain} | H_{t-1} = h)),$$

with transition matrix

$$\Pi = \begin{bmatrix} p(\text{pain}_t | \text{pain}_{t-1}) & p(\text{no pain}_t | \text{pain}_{t-1}) \\ p(\text{pain}_t | \text{no pain}_{t-1}) & p(\text{no pain}_t | \text{no pain}_{t-1}) \end{bmatrix}.$$

In healthy physiology, $p(\text{no pain}_t | \text{pain}_{t-1})$ is large: pain is expected to subside. In chronic pain, $p(\text{pain}_t | \text{pain}_{t-1})$ is high, encoding a maladaptive expectation of persistence.

The degree of that bias is what baseline severity indexes. A patient presenting with high z_{pain} can be read, in HMM terms, as one whose transition prior π already assigns high probability to persistence. The HMM does not by itself require that z enter the response surface in Chapter 4, that is a modeling choice motivated by the empirical literature on treatment effect heterogeneity, but it offers an account of why baseline severity should carry predictive weight beyond what a nuisance-covariate reading would suggest.

The emission model θ_h governs how internal states generate sensory signals:

$$\theta_h = (p(V_t = \text{noxious} \mid H_t = h), p(V_t = \text{harmless} \mid H_t = h)).$$

A well-calibrated emission model decodes harmless signals as harmless and noxious signals as noxious. In chronic pain, θ can shift so that ambiguous signals are read as noxious, supplying the persistence hypothesis with confirming evidence at the sensory level. The transition prior and the emission model can therefore both be miscalibrated, and the two failures reinforce one another. The chapter develops the transition prior further because that is the component that connects most directly to baseline severity and to the response surface in Chapter 4.

Recursive Posterior

The brain’s belief about H_t given all evidence so far follows from Bayes’ rule. Starting from

$$p(H_t \mid V_{1:t}) \propto p(V_t \mid H_t; \theta) p(H_t \mid V_{1:t-1}),$$

expanding the predictive prior by the law of total probability gives the recursive update:

$$p(H_t \mid V_{1:t}) \propto p(V_t \mid H_t; \theta) \sum_{H_{t-1}} p(H_t \mid H_{t-1}; \pi) p(H_{t-1} \mid V_{1:t-1}).$$

A chronic pain system with π biased toward persistence will maintain the pain hypothesis even when sensory input is ambiguous. A healthy system updates flexibly, allowing recovery when evidence indicates safety.

Ketamine’s hypothesized mechanism of action operates at this level (Corrigan et al., 2022): by modulating NMDA receptor activity, it may lower the effective precision of the transition prior, allowing the system to weight incoming safety signals more heavily. A more rigid transition prior has more capacity to be loosened in this way than a flexible one, so patients with higher baseline severity stand to gain more from successful recalibration. This is the same pattern the response surface in Chapter 4 parameterizes empirically through the slope matrix B .

2.2 Dirichlet Priors on Transition and Emission Probabilities

Each categorical distribution in the model, the transition model π_h and the emission model θ_h , is assigned a Dirichlet prior:

$$\pi_h \sim \text{Dirichlet}(\alpha_{\pi_h}), \quad \theta_h \sim \text{Dirichlet}(\alpha_{\theta_h}),$$

where each α is a vector of positive parameters that represent pseudocounts or prior experience. For the two-outcome case (e.g., pain/no pain or noxious/harmless), the parameter vector is $\alpha = (\alpha_1, \alpha_2)$.

Why a Dirichlet Prior?

The Dirichlet distribution is the conjugate prior of the categorical likelihood: prior and posterior belong to the same family, allowing analytic updating. If $\theta = (\theta_1, \dots, \theta_K)$ are the category probabilities with Dirichlet prior $p(\theta \mid \alpha) \propto \prod_c \theta_c^{\alpha_c - 1}$, then after observing counts $k = (k_1, \dots, k_K)$ the posterior is:

$$\theta \mid k \sim \text{Dirichlet}(\alpha + k).$$

The counts k are sufficient statistics: the posterior depends on the data only through how many times each category was observed. The update rule is additive:

$$\alpha_{\text{post}} = \alpha_{\text{prior}} + k, \tag{2.1}$$

with posterior mean

$$\mathbb{E}[\theta_c \mid k] = \frac{\alpha_c + k_c}{\sum_j (\alpha_j + k_j)}. \tag{2.2}$$

Both π_h and θ_h in the pain HMM follow this structure. Each pseudocount $\alpha_{h,j}$ represents how many prior instances the brain has experienced of transitioning from state h to j , or of generating sensory category j from state h . Large $\alpha_{h,j}$ implies high prior confidence; small $\alpha_{h,j}$ allows flexible updating.

Prior Precision and Clinical Interpretation

It is convenient to express the Dirichlet parameters as a mean vector p and a precision n :

$$\alpha = n \cdot p, \quad p = (p_1, \dots, p_K), \quad \sum_c p_c = 1.$$

Here p encodes the expected probability of each outcome and n represents the strength of that expectation. After observing counts k :

$$\mathbb{E}[\theta_c | k] = \frac{np_c + k_c}{n + N}. \quad (2.3)$$

If n is large, the prior dominates and new evidence has little effect; if n is small, beliefs shift easily with new data.

This parameterization has clinical interpretation. A person who frequently experiences transient pain followed by recovery has large n with high prior weight on $p(\text{no pain} | \text{pain})$: strong confidence that pain resolves. A person with a long history of persistent pain has large n with $p(\text{pain} | \text{pain})$ near one: a rigid prior that pain continues. In both cases, n is large; what differs is whether the prior concentrates on recovery or on persistence, not the confidence with which it does so. New sensory episodes contribute counts k that update α , but when n is large relative to N , these updates are small. This is why patients with longer symptom histories are harder to recalibrate, accumulated prior precision resists correction by new evidence.

This accumulation also offers a mechanistic reading of the clinical monotonicity prior used in Chapter 4, where the slope coefficients linking baseline severity to treatment response are constrained to be negative. A more rigid transition prior has more room to move under an effective intervention, which is consistent with the empirical pattern, documented in Voute et al. (2023), that motivates the constraint directly. The HMM is one way to understand why such a pattern would arise.

The transition prior π becomes overly precise in chronic pain, assigning excessive probability to persistence while sensory likelihood θ is underweighted. Recovery corresponds to recalibrating this balance, helping restore the brain’s ability to infer safety when evidence indicates it.

Conclusion

The update $\alpha_{\text{post}} = np + k$ formalizes learning as the Bayesian accumulation of experience. Baseline severity indexes the accumulated prior precision that determines how readily the inferential system can be recalibrated by new evidence such as ketamine. This is why differences in the distribution of z between source and target populations are not merely demographic. The difference in mean baseline severity between populations, formalized in Chapter 4, represents differences in how readily two populations’ inferential

systems can be recalibrated, which is what gives it clinical meaning beyond a simple difference in means.

Chapter 3

A Causal Framework for Data Fusion

3.1 Problem Framing

The inferential goal established in Chapter 1 is to characterize how ketamine treatment changes chronic pain and depression outcomes in TGD populations, where no trial evidence exists. The two available studies, Pickering et al. (2025) and Tabernacki et al. (2025), are each insufficient on their own: the ketamine cohort identifies a treatment response in a non-TGD population, and the TGD epidemiologic study characterizes the target population without any treatment exposure. This absence of trial evidence in the target population creates a structural identification problem: the available data are not sufficient to uniquely determine how ketamine would affect TGD patients, but they are sufficient to bound the answer. The framework developed in this chapter highlights those bounds, but bounding is a different problem than standard transportability settings, where individual-level data exist in both populations.

In an ideal setting, the causal effect of ketamine in this population would be estimated by analyzing a dataset in which TGD patients with chronic pain receive ketamine and are followed over time. No such dataset exists. However, there are two separate published studies that contain partial information to make such an inference: Pickering et al. (2025), a prospective multicenter observational cohort of $n = 585$ chronic pain patients receiving ketamine over twelve months, whose sample is not described as TGD and is

treated here as a predominantly cisgender source population; and Tabernacki et al. (2025), a large epidemiologic study characterizing prevalence, severity and psychiatric comorbidity in adults with TGD, which does not contain ketamine exposure.

The strategy of this chapter is to fuse these two studies using Pearl’s transportability framework (Bareinboim and Pearl, 2016). Fusion is permissible only if the two studies jointly identify the target estimand. Section 3.4 establishes this identification result using selection diagrams and S -admissibility; the upcoming section motivates the problem and introduces notation.

Notation

$S \in \{0, 1\}$ denotes population membership: $S = 0$ for the ketamine source cohort, $S = 1$ for the TGD target population. $Z = (Z_{\text{pain}}, Z_{\text{dep}})$ denotes the random vector of baseline pain and depression severity; z denotes a realized value. $\Delta = (\Delta_{\text{pain}}, \Delta_{\text{dep}})$ denotes 12-month change under ketamine, defined as $Y_{12\text{m}} - Y_{\text{baseline}}$; negative values indicate improvement. $do(\text{ketamine} = 1)$ denotes Pearl’s intervention operator (Pearl, 2009): $P(Y \mid do(X = x))$ is the distribution of Y that would be observed if X were set to x by an external intervention, distinct from the conditional $P(Y \mid X = x)$, which is the distribution of Y among units that happened to have $X = x$. $P^*(\cdot) \equiv P(\cdot \mid S = 1)$ denotes probabilities in the target population. The primary estimand of the thesis is $\psi = (\psi_{\text{pain}}, \psi_{\text{dep}})$, the bivariate transported mean treatment response in the TGD population; it is defined formally in Chapter 4.

The choice of Z as the conditioning set is substantive. As developed in Chapter 2, baseline pain and depression severity index the accumulated prior precision of the underlying inferential system, which governs how readily that system can be recalibrated by an effective intervention. Differences in Z between source and target populations therefore carry mechanistic content beyond a simple demographic shift, and the difference in mean baseline severity between populations that drives the transport adjustment in Chapter 4 inherits clinical meaning from this account.

3.2 A Simple Fusion Example

Before introducing the formal fusion of the ketamine, chronic pain, and TGD studies, it is helpful to look at a simple example where every step can be

computed by hand. The example below is a simpler version of the same problem, containing two studies, two populations, and a target effect that neither study can identify alone.

Setup

The example concerns two schools, A and B, evaluating whether a new teaching method improves exam pass rates. In this example X , Y , and Z are local to the toy setting and do not carry the same meaning as in the ketamine framework above; $P^*(\cdot)$ retains its meaning from the notation paragraph. Define:

$$\begin{aligned} X &= \begin{cases} 0 & \text{old teaching method} \\ 1 & \text{new teaching method} \end{cases} \\ Y &= \begin{cases} 0 & \text{fail} \\ 1 & \text{pass} \end{cases} \\ Z &= \begin{cases} 0 & \text{morning section} \\ 1 & \text{evening section} \end{cases} \end{aligned}$$

The Two Available Studies

No single dataset contains all of X , Y , and Z in the target population. Instead, we have two partial sources:

- **Study 1 (randomized trial, School A).** Students are randomly assigned to $X \in \{0, 1\}$. We observe (X, Z, Y) for all participants. Because assignment is random, we assume no confounding of X : the observed conditional $P(Y \mid X = x, Z = z)$ equals the interventional $P(Y \mid do(X = x), Z = z)$.
- **Study 2 (administrative data, School B).** We observe only Z , no teaching method is recorded, and no outcome is available. This study identifies the marginal covariate distribution in the target population:

$$P^*(Z) = P(Z \mid S = 1),$$

where $S = 1$ denotes School B membership. The selection variable S here plays the same role it will play throughout the chapter, indexing

population membership; the convention $S = 0$ for source and $S = 1$ for target is fixed in Section 3.3 for the ketamine setting. The toy example tells us nothing about how students respond to teaching.

We are interested in the effect of the new teaching method on exam pass rates at School B. The question is whether we can use School A’s information to identify this causal effect in a different population, a process called data fusion (Bareinboim and Pearl, 2016).

When Data Fusion Is Possible

Fusion is only justified when the two studies can be viewed as arising from the same underlying causal process. This assumption asserts that the way X and Z jointly influence Y is invariant across schools, even though the data were collected in different places. What is allowed to differ across populations in order to perform data fusion is the distribution of Z (e.g., School B may have more evening students than School A).

Following Pearl (2009) and Bareinboim and Pearl (2016), these two conditions, shared causal mechanism and shift only in Z , are encoded in a selection diagram. Population differences are represented with a selection node S : the arrow $S \rightarrow Z$ encodes that the distribution of Z may differ across populations, while the absence of a direct $S \rightarrow Y$ arrow encodes the invariance assumption that once Z is fixed, the outcome mechanism is the same in both schools.

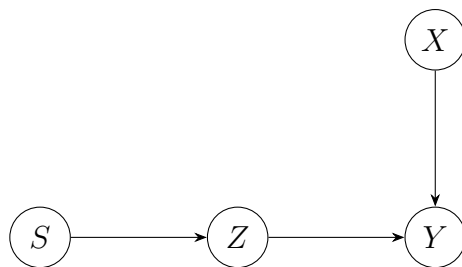


Figure 3.1: Selection diagram for the toy example (Bareinboim and Pearl, 2016). $S \rightarrow Z$ encodes population differences in covariate distribution; the absence of $S \rightarrow Y$ encodes the invariance assumption.

The Fusion Formula

Under the two assumptions above, identifying the effect in School B requires two quantities: how outcomes respond to X within each value of Z , captured by $P(Y \mid do(X = 1), Z = z)$, and how common each value of Z is in School B, captured by $P^*(Z = z)$, where $P^*(\cdot)$ denotes probabilities in the target population (School B). Study 1 provides the first and Study 2 provides the second.

The formula follows from two steps. First, apply the law of total probability over Z in the target population:

$$P^*(Y = 1 \mid do(X = 1)) = \sum_z P^*(Y = 1 \mid do(X = 1), Z = z) P^*(Z = z).$$

Second, apply the invariance assumption: since the outcome mechanism is the same in both schools once Z is fixed,

$$P^*(Y = 1 \mid do(X = 1), Z = z) = P(Y = 1 \mid do(X = 1), Z = z).$$

Substituting gives the fusion formula:

$$P^*(Y = 1 \mid do(X = 1)) = \sum_{z \in \{0,1\}} P(Y = 1 \mid do(X = 1), Z = z) P^*(Z = z).$$

The right-hand side is identified: $P(Y = 1 \mid do(X = 1), Z = z)$ comes from Study 1 and $P^*(Z = z)$ comes from Study 2.

Connection to the Ketamine Setting

The toy example has the same causal structure as the ketamine problem. In both cases, two studies each provide only part of the information needed:

- the ketamine cohort corresponds to Study 1, providing information about how pain and depression outcomes change under ketamine exposure, analogous to $P(Y \mid do(X), Z)$;
- the TGD chronic pain study corresponds to Study 2, providing the distribution of baseline characteristics in the target population $P^*(Z)$ but containing no ketamine exposure;

- the target quantity is the transported interventional effect of ketamine in the TGD population:

$$P^*(\Delta_{\text{pain}}, \Delta_{\text{dep}} \mid do(\text{ketamine})).$$

The ketamine study tells us what happens at each baseline severity profile when ketamine is given; the TGD study tells us how prevalent each profile is in the target population. Combining them allows us to estimate ketamine’s effect in a population where ketamine was never administered. The remainder of this chapter formalizes this structure using selection diagrams, S -admissibility, and Pearl’s transportability framework (Pearl, 2009; Bareinboim and Pearl, 2016).

3.3 Formal Fusion Goal

We now formalize the inferential goal in the ketamine setting. Let $\Delta = (\Delta_{\text{pain}}, \Delta_{\text{dep}})$ denote the 12-month change under ketamine, as defined in the notation paragraph above; negative values indicate improvement. In the ketamine cohort (source population, $S = 0$), Δ is observed under active ketamine treatment (Pickering et al., 2025). In the TGD chronic pain study (target population, $S = 1$), no ketamine is administered, so Δ is unobserved; we observe baseline covariates including pain severity, depression severity, and psychiatric comorbidity (Tabernacki et al., 2025).

Target Estimand

The causal quantity of interest is the joint interventional distribution

$$P^*(\Delta \mid do(\text{ketamine} = 1)),$$

that is, the distribution of pain and depression changes that would be observed if every individual in the TGD population received ketamine under the same clinical protocol as Pickering et al. (2025): intravenous subanesthetic ketamine administered in a chronic pain clinic setting over twelve months.

This estimand cannot be obtained from either dataset alone. The ketamine cohort does not represent the TGD population: Pickering et al. (2025) recruited chronic pain patients from French pain clinics and does not report TGD status; the sample is treated here as predominantly cisgender. The

TGD dataset (Tabernacki et al., 2025) contains no ketamine exposure and therefore no information about Δ .

Transport Covariates

Let $Z = (Z_{\text{pain}}, Z_{\text{dep}})$ denote baseline pain and depression severity. Pickering et al. (2025) reports these on continuous scales: NPRS (0–10) for pain and HADS (0–21) for depression, with baseline means, standard deviations, sex-stratified trajectories, and effect sizes for 12-month changes. These summaries constrain but do not uniquely determine $P(\Delta \mid \text{do}(\text{ketamine}), Z, S = 0)$. Tabernacki et al. (2025) reports age-adjusted incidence rates of chronic pain, hazard ratios for Major Depressive Disorder (MDD) and Generalized Anxiety Disorder (GAD) on chronic pain incidence, and demographic summaries, but no continuous severity scores on the NPRS or HADS scales.

The target population means of Z_{pain} and Z_{dep} are therefore derived assumptions: the elevated prevalence and hazard ratios in Tabernacki et al. (2025) are taken as evidence that TGD adults have higher baseline severity than the source cohort, and the target distribution $P^*(Z)$ is specified accordingly. This is a modeling assumption and is flagged as such in Chapter 4.

The ketamine cohort identifies the conditional interventional response

$$P(\Delta \mid \text{do}(\text{ketamine}), Z, S = 0),$$

while the TGD dataset identifies, up to this modeling assumption, the marginal distribution

$$P^*(Z) = P(Z \mid S = 1).$$

Source-Target Causal Structure

The two studies are linked through a selection variable $S \in \{0, 1\}$, where $S = 0$ denotes the ketamine source cohort and $S = 1$ the TGD target population. Following Pearl (2009) and Bareinboim and Pearl (2016), population differences are encoded in the selection diagram shown in Figure 3.2. The arrow $S \rightarrow Z$ encodes that the distribution of baseline covariates may differ across populations; the absence of a direct $S \rightarrow \Delta$ arrow encodes the assumption that once Z is fixed, the ketamine response mechanism is the same in both populations. This is the identifying assumption of the framework.

After intervening on ketamine, the only path from S to Δ in the diagram is $S \rightarrow Z \rightarrow \Delta$. Conditioning on Z blocks this path, meaning that once

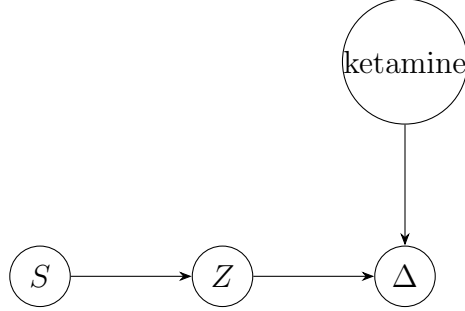


Figure 3.2: Selection diagram for ketamine fusion (Bareinboim and Pearl, 2016). $S \rightarrow Z$ encodes population differences in covariate distribution; the absence of $S \rightarrow \Delta$ encodes the invariance assumption.

Z is fixed, knowing which population an individual came from provides no additional information about Δ . Z is therefore S -admissible; the formal derivation of what this implies is given in Section 3.4.

Implication for Fusion

Under S -admissibility, the target interventional distribution decomposes as:

$$P^*(\Delta \mid do(\text{ketamine})) = \int P(\Delta \mid do(\text{ketamine}), Z = z, S = 0) P^*(Z = z) dz. \quad (3.1)$$

This expresses the TGD population’s ketamine response as an average of source-population responses at each baseline severity profile, weighted by how prevalent each profile is in the TGD population.

Chapter 4 estimates the mean of this distribution: the expected change in pain and depression under ketamine in the TGD population, constructed by integrating the source response surface over the target covariate distribution. The remainder of this chapter justifies equation (3.1) using selection diagrams and do-calculus.

3.4 Pearl’s Data Fusion Framework and Causal Structure

The derivation in this section follows from the transportability framework of Pearl (2009) and the data fusion formalization of Bareinboim and Pearl (2016).

The selection diagram and S -admissibility condition established in Section 3.3 justify the transport decomposition informally. This section derives equation (3.1) formally using Pearl’s do-calculus (Pearl, 2009).

S -Admissibility: Definition and Justification

What turns our selection diagram into a transport formula is S -admissibility. Z is S -admissible if, once we condition on Z and intervene on ketamine, the outcome Δ no longer depends on which population an individual came from. In that case, the source and target populations share a common conditional response surface $P(\Delta \mid do(ketamine), Z)$, and differences between them are exclusively compositional, via $P(Z \mid S)$.

Graphical definition. Let $G_{do(ketamine)}$ denote the graph obtained by deleting all arrows pointing into the ketamine node; this is the manipulated graph, representing the intervention structurally so that ketamine is set externally rather than by its usual causes. Because ketamine has no parents in the original selection diagram, as patients in Pickering et al. (2025) were referred to pain clinics and received ketamine as standard care, with no modeled determinants of who received it, $G_{do(ketamine)}$ is identical to Figure 3.2. The only path from S to Δ is $S \rightarrow Z \rightarrow \Delta$. Conditioning on Z blocks this path. Once Z is fixed, no information about population membership S can reach Δ through any active path. Z is therefore S -admissible (Bareinboim and Pearl, 2016), which implies:

$$P(\Delta \mid do(ketamine), Z, S = 0) = P(\Delta \mid do(ketamine), Z, S = 1).$$

The only remaining difference between populations lies in $P(Z \mid S)$. The graphical condition establishes S -admissibility formally; the following assumptions state what must hold substantively for the transport formula to be valid.

Identification Assumptions

The transport formula rests on four assumptions.

1. ***S*-admissibility.** Once baseline severity Z is fixed, the ketamine response mechanism is invariant across populations. Ketamine acts through NMDA receptor antagonism and downstream glutamatergic modulation (Corrigan et al., 2022; Voute et al., 2023), a pathway with no known interaction with gender identity; differential response would require a biological mechanism linking gender identity to NMDA receptor function that is not established in the literature. Consistent with this, Tabernacki et al. (2025) reports hazard ratios for MDD on chronic pain incidence that are similar across TGD and cisgender cohorts, suggesting that the mechanism linking baseline comorbidity to pain outcomes operates similarly across populations. This assumption is clinically plausible but ultimately untestable without outcome data from TGD patients receiving ketamine. If TGD patients respond differently at the same baseline severity for reasons not captured by Z , the transported estimates inherit that bias.
2. **No residual confounding in source.** Within the source population, ketamine assignment is approximately independent of Δ conditional on Z . Patients in Pickering et al. (2025) were referred to pain clinics and received ketamine as standard care, limiting but not eliminating the possibility of unmeasured confounders such as treatment history or clinician prescribing patterns. If sicker or more treatment-refractory patients were preferentially selected for ketamine in ways not captured by Z , the source response surface would overstate the effect attributable to ketamine alone, and the transported estimate ψ would inherit this bias in the direction of larger apparent improvement.
3. **Instrument equivalence.** The covariate vector Z measures the same construct on the same scale in both populations. This assumption is weakened by the fact that Pickering et al. (2025) and Tabernacki et al. (2025) use different instruments for pain and depression severity; if they measure different aspects of the same construct, the implied difference in mean baseline severity between populations could be systematically wrong.

4. **Positivity.** For every covariate value z in the support of $P^*(Z)$, there exist source individuals with comparable baseline profiles. The target means and source means used in the primary analysis, specified in Chapter 4, are within one to two standard deviations of each other, making violations of this assumption unlikely.

Do-Calculus Rules

Pearl’s do-calculus consists of three rules governing the manipulation of interventional and observational distributions (Pearl, 2009). The derivation in Section 3.5 requires only Rule 2; Rules 1 and 3 are stated here for completeness but are not invoked.

- **Rule 1 (Insertion/Deletion of Observations):** if a variable is conditionally independent of the outcome given the existing conditioning set and the intervention, it can be freely added to or removed from the conditioning set without changing the distribution.
- **Rule 2 (Action/Observation Exchange):** when an intervention on X does not open any new paths to Y given Z , intervening and conditioning are equivalent:

$$P(Y \mid do(X = x), Z) = P(Y \mid X = x, Z).$$

- **Rule 3 (Insertion/Deletion of Actions):** when an intervention on X has no causal effect on Y given Z , the intervention can be added or removed.

3.5 Fusion Formula and Modeling Strategy

Formal Derivation of the Transport Formula

Now, we derive equation (3.1). Recalling that $P^*(\cdot) \equiv P(\cdot \mid S = 1)$:

$$P^*(\Delta \mid do(ketamine)) = P(\Delta \mid do(ketamine), S = 1).$$

Apply the law of total probability over Z (written as an integral for continuous Z , as in the ketamine setting):

$$\begin{aligned} P(\Delta \mid do(\text{ketamine}), S = 1) &= \int P(\Delta \mid do(\text{ketamine}), Z = z, S = 1) \\ &\quad \times P(Z = z \mid do(\text{ketamine}), S = 1) dz. \end{aligned} \quad (3.2)$$

Because Z is a pre-treatment covariate with no descendant of ketamine as a parent, intervening on ketamine does not change the distribution of Z . The intervention can therefore be dropped from the weight:

$$P(Z = z \mid do(\text{ketamine}), S = 1) = P(Z = z \mid S = 1) = P^*(Z = z).$$

Substituting into (3.2):

$$P^*(\Delta \mid do(\text{ketamine})) = \int P(\Delta \mid do(\text{ketamine}), Z = z, S = 1) P^*(Z = z) dz. \quad (3.3)$$

The conditional $P(\Delta \mid do(\text{ketamine}), Z = z, S = 1)$ is not observed in the target population. By S -admissibility, we swap $S = 1$ for $S = 0$:

$$P(\Delta \mid do(\text{ketamine}), Z = z, S = 1) = P(\Delta \mid do(\text{ketamine}), Z = z, S = 0).$$

Finally, by Rule 2 of do-calculus, since ketamine has no parents in $G_{do(\text{ketamine})}$, intervening and conditioning are equivalent in the source population:

$$P(\Delta \mid do(\text{ketamine}), Z = z, S = 0) = P(\Delta \mid \text{ketamine}, Z = z, S = 0).$$

This step requires that ketamine assignment is unconfounded given Z within the source population; this is Assumption 2 of Section 3.4.

Substituting gives the transport formula:

$$P^*(\Delta \mid do(\text{ketamine})) = \int P(\Delta \mid \text{ketamine}, Z = z, S = 0) P^*(Z = z) dz,$$

which is equation (3.1). $P(\Delta \mid \text{ketamine}, z, S = 0)$ is identified from the ketamine cohort; $P^*(Z)$ is identified from the TGD study.

The transport formula (3.1) implies a specific factorization of the joint distribution over (Z, Δ) in the target population under ketamine. Any model compatible with (3.1) must satisfy:

$$P^*(Z, \Delta \mid do(\text{ketamine})) = P^*(Z) P(\Delta \mid \text{ketamine}, Z, S = 0). \quad (3.4)$$

This factorization is a consequence of the causal structure established in Section 3.4. It requires two components to be specified separately: a model for $P^*(Z)$, the target population covariate distribution, and a model for $P(\Delta \mid \text{ketamine}, z, S = 0)$, the source-population conditional response mechanism. These two components come from different studies and are combined only at the level of the transport formula, not by pooling the datasets directly.

From Formula to Generative Scheme

Any joint distribution written as a product of a marginal and a conditional can be sampled in two steps: draw from the marginal, then draw from the conditional given that draw. The factorization above has this form, yielding a generative scheme: draw a covariate profile $Z^{(i)} \sim P^*(Z)$, then draw a response $\Delta^{(i)} \sim P(\Delta \mid \text{ketamine}, Z^{(i)}, S = 0)$. Repeating this for $i = 1, \dots, N$ produces a collection whose distribution approximates the target interventional distribution. This is the structure implemented in Chapter 4: $P(\Delta \mid \text{ketamine}, Z, S = 0)$ is parameterized as a bivariate linear response surface constrained to match the published source summaries, and $P^*(Z)$ is specified from Tabernacki et al. (2025) up to the modeling assumption described in Section 3.3. The constrained prior predictive Monte Carlo procedure then approximates the outer integral over $P^*(Z)$.

3.6 Conclusion

This chapter established the causal foundation for fusing the ketamine cohort and the TGD chronic pain study. Beginning with a toy example, we illustrated how interventional effects can be transported across populations when two pieces of information are available: a conditional response mechanism identified in a source study, and the marginal distribution of relevant covariates in a target population. We then showed that the ketamine–TGD problem has the same structure.

Using Pearl’s transportability framework (Pearl, 2009; Bareinboim and Pearl, 2016), selection diagrams, S -admissibility, and Rule 2 of do-calculus, we derived the transport formula:

$$P^*(\Delta \mid \text{do}(\text{ketamine})) = \int P(\Delta \mid \text{ketamine}, Z = z, S = 0) P^*(Z = z) dz.$$

Graphical separation in the manipulated graph established S -admissibility, which swapped the population index from $S = 1$ to $S = 0$; Rule 2 replaced the intervention with an observable conditional; and the observational nature of Pickering et al. (2025) was flagged as an identifying assumption rather than a consequence of study design.

The formula implies a factorization of the joint distribution over (Z, Δ) in the target population:

$$P^*(Z, \Delta \mid do(\text{ketamine})) = P^*(Z) P(\Delta \mid \text{ketamine}, Z, S = 0),$$

The framework rests on four maintained assumptions, stated formally in Section 3.4: S -admissibility of Z , no residual confounding of ketamine within the source population conditional on Z , instrument equivalence across studies, and positivity of the covariate overlap.

The transportability framework developed in this chapter addresses a setting that is structurally harder than the standard case in the literature. Most applied transportability work assumes access to individual-level data in both the source and target populations. Here, only summary covariate information is available from the target population: marginal prevalence estimates and hazard ratios, not individual-level covariate profiles.

This restriction means that the transport formula is identified but the quantities inside it cannot all be computed from the published summaries alone. Chapter 4 addresses the gap by parameterizing the response surface and characterizing the residual uncertainty that the published data leave unresolved.

The value of this framework lies in making the conditions for transportability explicit and falsifiable. Each assumption specifies a structural claim about the relationship between the source and target populations, and the identification result holds only to the extent that those claims hold. When an assumption is violated, whether through measurement mismatch or unobserved effect modifiers, the framework identifies which assumption failed and what consequence that failure has for the transported estimate.

With the identification conditions and the transport formula in place, the remaining question is whether the available data are sufficient to evaluate it. Chapter 4 operationalizes the factorization developed in this chapter: the source response mechanism is parameterized as a bivariate linear response surface constrained to match the published summaries from Pickering et al. (2025), and the target covariate distribution is specified from Tabernacki

et al. (2025). Constrained prior predictive Monte Carlo then approximates the outer integral, producing the transported mean effect ψ that is the primary estimand of the thesis.

Chapter 4

Computing the Transported Estimand

Chapter 3 established that the transported mean treatment response in the TGD population is identified under structural invariance and S -admissibility. Identification, however, involves distributions. It says that the transported mean treatment response is in principle expressible as a functional of observable quantities across the two studies. It does not say how to compute it, nor does it say what to do when the required functional cannot be evaluated directly from what the studies actually report. The chapter addresses both problems. Section 4.1 bridges the integral identity from Chapter 3 to a parametric response surface that can be computed in practice. Section 4.2 defines the transported mean treatment response ψ as a scalar estimand. Section 4.3 shows why analytic computation fails given only published summaries. The remaining sections develop the generative model, the Monte Carlo procedure, and a sensitivity analysis over the structural quantities that remain unidentified after all published constraints are applied.

The central finding of the chapter is that the transported estimate is statistically indistinguishable from the source mean response: uncertainty about how baseline severity predicts treatment response is an order of magnitude larger than the transport adjustment itself. This near-null result reveals information about what the available data can and cannot support, and the remainder of the chapter explains why.

4.1 From Identification to Computation

This section translates the identification result of Chapter 3 to a computable parametric response surface, and identifies the residual uncertainty that the rest of the chapter addresses. The transport formula derived in Chapter 3 expresses the target interventional distribution as

$$P^*(\Delta \mid do(\text{ketamine})) = \int P(\Delta \mid \text{ketamine}, Z = z, S = 0) P^*(Z = z) dz. \quad (4.1)$$

The transported mean response ψ , defined formally in Section 4.2, is the scalar summary of this distribution that the rest of the chapter aims to compute. Two components are required, the conditional response distribution $P(\Delta \mid \text{ketamine}, z, S = 0)$ from the source population, and the marginal covariate distribution $P^*(Z)$ from the target population. Neither is fully available from the published literature.

The summaries constrain the marginal distributions of Z and Δ separately, but they do not identify $P(\Delta \mid z, S = 0)$ as a function over z . Section 4.3 shows why this holds. Marginal moments cannot recover conditional relationships when the parameters governing those relationships are unconstrained by the available data. In particular, the slope matrix B governing how baseline severity predicts treatment response is not recoverable from published moments alone. Without B , the integral in equation (4.1) cannot be evaluated.

From Pickering’s model to a parametric response surface

Pickering et al. (2025) analyze the source cohort using a linear mixed-effects model. They do not formally define the model, but the published description, with time as a fixed effect, patient as a random effect, corresponds to a Laird-Ware specification (Laird and Ware, 1982) of the form

$$Y_{it} = \beta_0 + \beta_1 t + u_i + \epsilon_{it}, \quad (4.2)$$

where Y_{it} is the outcome score (pain or depression) for patient i at time t , β_0 is the population-average baseline, β_1 is the population-average rate of change, $u_i \sim \mathcal{N}(0, \tau^2)$ is the patient-specific random intercept capturing

each patient’s stable deviation from the population average, and ϵ_{it} is residual noise.

Our outcome of interest is the 12-month change, $\Delta_i = Y_{i,12} - Y_{i,0}$. Substituting equation (4.2) at $t = 12$ and $t = 0$ and taking the difference,

$$\begin{aligned}\Delta_i &= (\beta_0 + 12\beta_1 + u_i + \epsilon_{i,12}) - (\beta_0 + u_i + \epsilon_{i,0}) \\ &= 12\beta_1 + (\epsilon_{i,12} - \epsilon_{i,0}).\end{aligned}$$

The patient random effect u_i cancels. Δ_i is therefore a fixed-effects quantity in Pickering’s model. It has no patient-level random component, only a population-level slope and mean-zero residual noise. This is what justifies treating Δ as the response variable in a fixed-effects regression on baseline covariates, the random effect that complicates the original model of Y_{it} does not appear in the model of Δ_i .

Pickering’s specification has β_1 as a single population-level constant: every patient is modeled as changing at the same expected rate. Our extension is to allow β_1 to depend on baseline covariates z , since covariate shift between source and target can only enter the transported estimate through some such dependence. Writing $\beta_1(z)$ and taking the conditional expectation,

$$E[\Delta_i \mid z] = 12 \beta_1(z). \quad (4.3)$$

We adopt a linear functional form for $\beta_1(z)$, consistent with outcome regression approaches to transportability (Dahabreh et al., 2019). Throughout what follows we work with baseline covariates standardized to source-population units,

$$\tilde{z} = (z - \mu_Z) \oslash \sigma_Z,$$

where $\oslash$ denotes element-wise division. The response surface is then

$$E[\Delta_i \mid \tilde{z}, S = 0] = \alpha + B\tilde{z}, \quad (4.4)$$

where $\alpha \in \mathbb{R}^2$ is the expected response at the source-population mean baseline ($\tilde{z} = 0$) and $B \in \mathbb{R}^{2 \times 2}$ collects the slope coefficients in source-standardized units. Standardization is a reparameterization choice, not an additional assumption. Standardization fixes the intercept to a clinically interpretable point and simplifies the anchor constraint derived below.

The transport formula (4.1) requires the inner expectation $P(\Delta \mid \text{ketamine}, S = 0)$ as a function of z , but Pickering’s analysis does not estimate that function directly. Equation (4.4) supplies it as a parametric extension of Pickering’s

mixed-effects framework. The slope matrix B in that extension, however, is not identified from Pickering’s marginal summaries, which is what motivates the simulation procedure developed in the rest of the chapter.

The second component in the transport formula is the target covariate distribution $P^*(Z)$. The TGD epidemiologic study Tabernacki et al. (2025) reports the prevalence and relative risk of chronic pain and depression in TGD adults, stratified by gender identity and hormone therapy status. It does not report continuous symptom severity scores on the NPRS or HADS scales used in the source cohort. Two consequences follow.

First, gender identity and hormone therapy status are not included in Z . These variables are not observed in the source cohort, so including them would require extrapolating the response surface beyond the support of the source data and undermine the identification argument. Sex, which appeared as a covariate in earlier versions of this work, is excluded for the same reason: the source cohort’s binary sex categories do not map cleanly onto a TGD target population, and the chapter does not attempt to bridge the categorical mismatch.

Second, the target marginal means μ^* and standard deviations σ^* in baseline pain and depression cannot be read directly from (Tabernacki et al., 2025). The reference values $\mu^* = (6.2, 9.1)$ used in the primary analysis are working assumptions, chosen to sit modestly above the source cohort means (4.8, 6.9) on the basis that TGD adults carry elevated baseline burden of pain and depression as documented in (Tabernacki et al., 2025).

The specific reference values are not themselves identified from the data, and the sensitivity of ψ to this choice is investigated systematically in the μ^* sweep of Section 4.6.

Addressing that gap is the purpose of the present chapter. Rather than computing a single value of ψ , we characterize the distribution of ψ values consistent with all published constraints. The identification result of Chapter 3 remains intact. But, instead of a point estimate, we obtain a constrained prior predictive distribution over ψ that reflects what is consistent with the available evidence.

4.2 The Estimand

The target of inference is a scalar summary of the transported interventional distribution defined in equation (4.1). Rather than characterizing the full

distribution $P^*(\Delta \mid do(\text{ketamine}))$, we focus on its mean. Because the mean of an identified distribution is itself identified, ψ inherits its identification status directly from the Chapter 3 transport formula; the simplification to the mean changes the target of inference without introducing a new identification assumption. We denote this scalar target

$$\psi = E_{z \sim P^*(z)}[E[\Delta \mid \text{ketamine}, z, S = 0]] = \int E[\Delta \mid \text{ketamine}, z, S = 0] P^*(z) dz. \quad (4.5)$$

Equation (4.5) follows directly from Pearl’s transportability framework Pearl (2009), in which the transported interventional distribution in a target population is expressed as an integral of the source-population conditional response against the target covariate distribution. The data fusion formalization of the transported interventional distribution (the distribution of ketamine treatment outcomes in the target TGD population), $P^*(\Delta \mid do(\text{ketamine}))$, and the conditions under which it is identified from multisource data, are developed in Bareinboim and Pearl (2016) and applied to the ketamine–TGD setting in Chapter 3. Here $\Delta = (\Delta_{\text{pain}}, \Delta_{\text{dep}})$ is the bivariate vector of 12-month changes in pain and depression under ketamine, $z = (z_{\text{pain}}, z_{\text{dep}})$ is the vector of baseline covariates, and $P^*(z)$ is the target TGD population covariate distribution. Because Δ is bivariate, $\psi = (\psi_{\text{pain}}, \psi_{\text{dep}})$ is itself a two-dimensional quantity, and the computation proceeds separately for each component.

Defining ψ

ψ is the transported mean treatment response, the expected change in pain and depression that would be observed if the ketamine protocol from the source cohort were administered to a randomly drawn member of the TGD target population. It answers the question of given what we know about how ketamine works in the source population and what we know about baseline symptom burden in TGD adults, what mean improvement should we expect?

The outer expectation over $z \sim P^*(z)$ in equation (4.5) integrates over the target covariate distribution, which encodes the fact that the TGD population differs systematically from the source cohort in its baseline pain and depression levels. The inner expectation averages Δ over the response distribution in the source population at each fixed baseline profile z . The covariate shift motivates the transportability framework in Chapter 3: the source pop-

ulation and the target population share a response mechanism, but they differ in who presents for treatment.

What ψ is not

ψ is not an average treatment effect in the causal contrast sense. An average treatment effect compares potential outcomes under treatment and control: $E[Y(1) - Y(0)]$. No such comparison is available here. The ketamine cohort contains no untreated control arm from which to estimate counterfactual outcomes, and the TGD study contains no treatment exposure at all. ψ is therefore better understood as a transported mean response rather than a transported causal contrast. It quantifies expected improvement under treatment, not improvement relative to a no-treatment baseline.

The distinction matters for interpretation. A large negative ψ tells us that if TGD patients received ketamine under the same protocol used in Corrigan et al. (2022), we would expect substantial reductions in pain and depression scores on average. It does not tell us how much of that reduction is attributable to ketamine specifically versus natural symptom trajectory, placebo response, or regression to the mean. Those questions require a control condition that neither study provides.

While ψ does not subtract a control-group baseline, it remains the relevant quantity for clinical decision-making in a single-arm context: it provides a projected mean outcome for a population currently unobserved in ketamine trials, and it is precisely the quantity a clinician or policy-maker would need to assess whether initiating a ketamine protocol in TGD patients is warranted given the available evidence.

Connection to the transport formula

Equation (4.5) is the mean of the interventional distribution established in Chapter 3. Every step of the identification argument applies directly: the inner expectation $E[\Delta \mid \text{ketamine}, z, S = 0]$ is identified from the source cohort under the S -admissibility condition, and $P^*(z)$ is identified from the target study. The only new element introduced here is the restriction to the mean rather than the full distribution, which simplifies computation without changing the identification logic.

4.3 From Identity to Simulation

Section 4.1 introduced the linear response surface $E[\Delta \mid \tilde{z}, S = 0] = \alpha + B\tilde{z}$ as a parametric extension of Pickering’s mixed-effects model. Under linearity, the integral defining ψ reduces to a single matrix-vector product. The reduction is exact, but it depends on a slope matrix B that the published source summaries do not identify.

The analytic reduction under linearity

Under the linear response surface $E[\Delta \mid \tilde{z}, S = 0] = \alpha + B\tilde{z}$, where $\tilde{z} = (z - \mu_Z) \oslash \sigma_Z$ denotes baseline covariates standardized to source-population units, the integral defining ψ reduces to a single matrix-vector product. Starting from equation (4.5):

$$\psi = \int E[\Delta \mid z, S = 0] P^*(z) dz.$$

Substitute the linear response surface:

$$\psi = \int (\alpha + B(z - \mu_Z) \oslash \sigma_Z) P^*(z) dz.$$

Split the integral across the sum:

$$\psi = \int \alpha P^*(z) dz + \int B(z - \mu_Z) \oslash \sigma_Z P^*(z) dz.$$

Since α does not depend on z and $\int P^*(z) dz = 1$, and since B , μ_Z , and σ_Z are constant with respect to z :

$$\psi = \alpha + B \left(\int z P^*(z) dz - \mu_Z \right) \oslash \sigma_Z.$$

The remaining integral is the definition of $E_{P^*}[z] = \mu^*$. Writing $\mu_{\text{std}}^* = (\mu^* - \mu_Z) \oslash \sigma_Z$ for the target mean in source-standardized units:

$$\psi = \alpha + B \mu_{\text{std}}^*. \tag{4.6}$$

The expression is exact under linearity: ψ is fully determined by the intercept, the slope matrix, and the standardized target population mean. The linear response surface used here was derived in Section 4.1 as a parametric extension of Pickering’s mixed-effects framework; the sensitivity of ψ to uncertainty in B is addressed in Section 4.5.

Why B is not identified

B is not identified. Recovering a 2×2 slope matrix requires individual-level pairs $(\tilde{z}_i, \Delta_i)$ from the source population, from which the conditional relationship between baseline severity and treatment response can be estimated. The ketamine cohort provides no such pairs. It reports only marginal summaries: mean changes μ_Δ , baseline means μ_Z , and baseline standard deviations σ_Z . As established in Section 4.1, Pickering’s linear mixed-effects model fits time as the fixed effect rather than baseline severity, so the source analysis does not identify B .

Under the standardized parameterization, a source patient at the mean baseline corresponds to $\tilde{z} = 0$, so the published mean change satisfies

$$\mu_\Delta = E[\Delta \mid S = 0] = \alpha + B \cdot 0 = \alpha. \quad (4.7)$$

The intercept is therefore pinned to the published source mean change regardless of which B is drawn from the prior:

$$\alpha = \mu_\Delta. \quad (4.8)$$

This is the anchor constraint. The marginal moments fix α but say nothing about B , which has four free entries that no published quantity constrains.

The anchor constraint is what makes the simulation constrained rather than a free prior predictive Gabry et al. (2019): any slope matrix B drawn from the prior is paired with an intercept that exactly reproduces the published source mean change at the source mean baseline. The simulation procedure developed in Section 4.5 exploits this structure: B is drawn from a prior, α is set to μ_Δ , and ψ is evaluated for each draw. The residual freedom in the problem is contained entirely in B .

Simulation Strategy

Because B is unidentified, ψ is not a single number but a set of values, one for each slope matrix consistent with the published constraints. Rather than reporting a single point estimate that would misrepresent the uncertainty, we characterize the distribution of ψ across the space of plausible slope matrices. The procedure places a prior on B , draws T slope matrices from that prior, computes $\psi^{(t)}$ for each draw via equation (4.6), and summarizes the resulting collection $\{\psi^{(t)}\}_{t=1}^T$. The output is a constrained prior predictive distribution

over ψ Gabry et al. (2019); Gelman et al. (2013): not a posterior, because no likelihood updates the prior on B , but not a free prior predictive either, because α is fixed by the anchor constraint. The prior on B itself reflects weak judgmental assumptions rather than calibration to data; sensitivity of ψ to the prior scale σ_β is addressed directly in Section 4.6.

4.4 The Generative Model

Computing the distribution of ψ across slope matrices consistent with the published constraints requires three modeling choices: a parametric form for the conditional response surface, a prior distribution over the unidentified slope matrix B , and a distributional form for the target covariate distribution $P^*(Z)$. Each choice is made at the minimal level of complexity justified by the available evidence, and each is subjected to sensitivity analysis in Section 4.6.

The prior on B

B is the slope matrix that governs how baseline severity predicts treatment response: it captures whether patients who are sicker at baseline improve more, less, or the same amount under ketamine. Formally, B is a 2×2 matrix whose rows index outcomes (pain, depression) and whose columns index baseline covariates (baseline pain, baseline depression). We allow all four entries to be nonzero, permitting cross-outcome effects: baseline depression may predict change in pain, and vice versa. The full matrix is clinically appropriate because pain and depression are substantially comorbid in chronic pain populations Aaron et al. (2025), and restricting B to be diagonal would impose an independence assumption unsupported by the literature.

The prior on B is specified entry-wise. Off-diagonal entries B_{12} and B_{21} are drawn independently from

$$B_{ij} \sim \mathcal{N}(0, \sigma_\beta^2), \quad i \neq j, \quad (4.9)$$

with $\sigma_\beta = 0.15$. The prior is centered at zero, reflecting ignorance about the direction of cross-outcome effects. Since $E[B_{ij}] = 0$ for $i \neq j$, off-diagonal entries contribute nothing to the mean of ψ ; they contribute symmetric widening to the prior predictive interval without shifting its center.

Diagonal entries B_{11} (baseline pain $\rightarrow$ change in pain) and B_{22} (baseline depression $\rightarrow$ change in depression) are subject to a clinical monotonicity

prior, that is, higher baseline severity is associated with larger absolute improvement, so the diagonal slopes are constrained to be negative. The pattern is well-documented in chronic pain and depression research, where patients with greater symptom burden at baseline show larger absolute reductions under active treatment, a combination of regression to the mean and genuine treatment response heterogeneity Voute et al. (2023). The Bayesian-inference account of pain developed in Chapter 2 offers a mechanistic reading of the same pattern: a transition prior with greater accumulated precision has more room to be recalibrated by an effective intervention, predicting larger absolute change for patients whose baseline severity reflects more rigid priors.

We implement the constraint as a truncated normal prior:

$$B_{ii} \sim \mathcal{N}(0, \sigma_\beta^2) \cdot 1[B_{ii} < 0], \quad i = 1, 2, \quad (4.10)$$

with the same $\sigma_\beta = 0.15$. The negative-half-normal distribution concentrates mass on slopes that are small in magnitude but reliably negative, with expected value $E[B_{ii}] = -\sigma_\beta \sqrt{2/\pi} \approx -0.12$. The value $\sigma_\beta = 0.15$ is not calibrated to any specific trial estimate and reflects the expectation that baseline severity has a modest rather than dramatic effect on treatment response. The sensitivity of ψ to σ_β is assessed directly in Section 4.6.

Because $\sigma_\beta = 0.15$ places substantial mass near zero, the prior explicitly allows for approximately homogeneous treatment response, scenarios in which baseline severity has negligible predictive power and ketamine works nearly identically regardless of who receives it. The width of the prior predictive interval for ψ therefore reflects not only uncertainty about the direction of effect heterogeneity but also the possibility that no meaningful heterogeneity exists.

The target covariate distribution

We model the target covariate distribution $P^*(Z)$ as a bivariate normal:

$$Z^* \sim \mathcal{N}(\mu^*, \Sigma^*), \quad (4.11)$$

where $\mu^* = (\mu_{\text{pain}}^*, \mu_{\text{dep}}^*)$ are the target population baseline means, Σ^* is the target covariance matrix with marginal standard deviations $(\sigma_{\text{pain}}^*, \sigma_{\text{dep}}^*)$ and correlation ρ^* .

As established in Section 4.1, none of the parameters are directly reported in Tabernacki et al. (2025). The epidemiologic study reports binary

prevalence and hazard ratios for chronic pain and depression diagnoses in TGD adults, not continuous severity scores on the NPRS or HADS scales used in the source cohort. The parameters of $P^*(Z)$ are therefore modeling assumptions, specified as follows.

The target covariance matrix has the form

$$\Sigma^* = \begin{pmatrix} (\sigma_{\text{pain}}^*)^2 & \rho^* \sigma_{\text{pain}}^* \sigma_{\text{dep}}^* \\ \rho^* \sigma_{\text{pain}}^* \sigma_{\text{dep}}^* & (\sigma_{\text{dep}}^*)^2 \end{pmatrix}. \quad (4.12)$$

The marginal standard deviations $(\sigma_{\text{pain}}^*, \sigma_{\text{dep}}^*) = (2.5, 2.8)$ are set slightly larger than the source cohort values $(2.2, 2.6)$, reflecting the expectation that a population not selected through a specialized chronic pain clinic will exhibit greater dispersion in symptom severity. The baseline correlation ρ^* between pain and depression in the target population is unidentified and is varied across $\rho^* \in \{-0.3, 0.0, 0.3, 0.6\}$ in the sensitivity analysis of Section 4.6. At each Monte Carlo iteration, M target individuals are drawn from $Z^* \sim \mathcal{N}(\mu^*, \Sigma^*)$.

Target baseline means and the sensitivity strategy

The baseline means μ^* are the primary sensitivity parameter for the target distribution. The reference values $\mu^* = (6.2, 9.1)$ are set modestly above the source cohort means $(4.8, 6.9)$ to reflect the elevated burden of chronic pain and depression documented in TGD populations Tabernacki et al. (2025). Because Tabernacki et al. (2025) reports binary prevalence and hazard ratios rather than continuous NPRS or HADS scores, these values cannot be derived directly from the target-population literature; the sensitivity sweep over μ^* , not the reference values themselves, is what carries the predominant analytic weight. We vary them across a clinically plausible grid: $\mu_{\text{pain}}^* \in \{5.8, 6.2, 6.6\}$ and $\mu_{\text{dep}}^* \in \{8.5, 9.1, 9.7\}$, yielding nine scenarios that span the range of plausible target population baseline burden. The reference scenario $(6.2, 9.1)$ is used for the primary analysis; the full grid is reported in Section 4.6.

The transport formula derived in Chapter 3 involves two estimable quantities: the source mean response μ_{Δ} , identified from Pickering et al. (2025), and the covariate shift $\mu^* - \mu_Z$, informed by Tabernacki et al. (2025). The remaining unknown is the slope matrix B . The strategy of this chapter is to anchor the estimable quantities from the published data, then sweep over the prior on B to characterize how sensitive ψ is to the unknown slope matrix. The resulting distribution over ψ is a constrained prior predictive check

Gabry et al. (2019): it asks what the transported estimate could look like under different assumptions about treatment effect heterogeneity, given the data that are available.

4.5 The Monte Carlo Procedure

The generative model of Section 4.4 defines a distribution over ψ implicitly: each draw of B from the prior yields a deterministic value of ψ via equation (4.6), so the distribution of ψ is determined by passing the prior on B through that equation. Because equation (4.6) is linear in B , the resulting distribution could in principle be computed analytically. In practice, the truncated normal prior on the diagonal entries of B and the bivariate normal draw over M target individuals make the analytic form cumbersome. We therefore approximate the distribution of ψ by Monte Carlo, drawing $T = 4,000$ slope matrices and computing $\psi^{(t)}$ for each.

The procedure has two nested loops reflecting two distinct sources of variation. The outer loop captures parameter uncertainty: each iteration draws a new slope matrix $B^{(t)}$ from the prior, representing a different hypothesis about how baseline severity predicts treatment response. The inner loop captures covariate heterogeneity: for each $B^{(t)}$, we draw $M = 10,000$ individuals from the target distribution $P^*(Z)$ and average the predicted responses across them, approximating the integral $\int E[\Delta \mid z, S = 0] P^*(z) dz$ by a sample mean. The two loops are necessary because $\psi^{(t)}$ depends on both which slope matrix governs the response surface and which individuals are drawn from the target population. Collapsing them into a single loop would confound parameter uncertainty with sampling variability.

Throughout the algorithm, $\hat{\delta}_m$ denotes the predicted individual change for simulated target individual m , the sample analogue of the random variable Δ used in the population-level expressions above. We assume throughout that $\sigma_{Z,k} > 0$ for each covariate k , so that the standardization step in line 8 is well-defined.

The procedure is summarized in Algorithm 1.

Algorithm 1: Constrained prior predictive Monte Carlo for ψ

Input: Source moments $\mu_\Delta, \mu_Z, \sigma_Z$; target moments μ^*, σ^*, ρ^* ;
prior width σ_β ; outer draws T ; inner draws M
Output: Prior predictive samples $\{\psi^{(t)}\}_{t=1}^T$

Set $\alpha \leftarrow \mu_\Delta$; // anchor constraint

for $t = 1$ **to** T **do**

Draw off-diagonal entries: $B_{12}^{(t)}, B_{21}^{(t)} \sim \mathcal{N}(0, \sigma_\beta^2)$;
Draw diagonal entries: $B_{11}^{(t)}, B_{22}^{(t)} \sim \mathcal{N}(0, \sigma_\beta^2) \cdot 1[\cdot < 0]$;

for $m = 1$ **to** M **do**

Draw $z_m^* \sim \mathcal{N}(\mu^*, \Sigma^*)$;
Standardize to source units: $\tilde{z}_m \leftarrow (z_m^* - \mu_Z) \oslash \sigma_Z$;
Compute $\hat{\delta}_m \leftarrow \alpha + B^{(t)} \tilde{z}_m$;

end

$\psi^{(t)} \leftarrow \frac{1}{M} \sum_{m=1}^M \hat{\delta}_m$;

end

return $\{\psi^{(t)}\}_{t=1}^T$

The standardization step in line 8 is critical. The operator $\oslash$ denotes element-wise division: $(z_m^* - \mu_Z) \oslash \sigma_Z$ produces a vector whose k th entry is $(z_{mk}^* - \mu_{Z_k})/\sigma_{Z_k}$, not a matrix inverse. The slope matrix B operates on covariates expressed in units of source population standard deviations from the source population mean. Target individuals must therefore be standardized using the source moments (μ_Z, σ_Z) , not the target moments. Standardizing by target moments would place all target individuals at a mean of zero on the target scale, erasing the covariate shift that the transport procedure is designed to exploit. The covariate shift enters the computation precisely through the fact that target individuals, when expressed in source-standardized units, have nonzero mean.

To see why source standardization is necessary, consider an example. A TGD patient with raw baseline pain score 6.2, the target population mean μ_{pain}^* , represents a typical TGD patient. Under target standardization, this patient maps to

$$\tilde{z} = \frac{6.2 - 6.2}{2.5} = 0, \quad (4.13)$$

and every average TGD patient maps to zero. The response surface then predicts

$$\hat{\delta} = \alpha + B \cdot 0 = \alpha = \mu_{\Delta}, \quad (4.14)$$

the source mean effect. The covariate shift between populations is erased, and ψ collapses to μ_{Δ} regardless of B . Under source standardization, the same patient maps to

$$\tilde{z} = \frac{6.2 - 4.8}{2.2} = 0.636, \quad (4.15)$$

placing them 0.636 source standard deviations above the source mean. The response surface then predicts

$$\hat{\delta} = \alpha + B \cdot 0.636, \quad (4.16)$$

which with negative diagonal entries in B yields a larger improvement than μ_{Δ} . The fact that the average TGD patient is sicker than the average source patient is encoded in the positive standardized value, and the negative slopes translate that excess severity into a larger predicted response. This is why $\mu^* > \mu_Z$ componentwise is the engine of the transport result, and why standardizing by target moments would silently eliminate it.

Output and interpretation

Quantiles of $\{\psi^{(t)}\}_{t=1}^T$ are reported as interval estimates for ψ . The intervals quantify uncertainty about the slope matrix B , not sampling uncertainty about a fixed parameter. A 95% interval $[\psi_{0.025}, \psi_{0.975}]$ should be read as: across the space of response surfaces consistent with the published constraints and the prior on B , 95% yield a transported mean response in that range.

Computational notes

Under linearity, the sample mean $\frac{1}{M} \sum_{m=1}^M \hat{\delta}_m$ converges to $\alpha + B^{(t)} \mu_{\text{std}}^*$ as $M \rightarrow \infty$, where $\mu_{\text{std}}^* = (\mu^* - \mu_Z) \oslash \sigma_Z$ is the target mean in source-standardized units. The inner loop is therefore approximating a quantity that could be computed in closed form. We retain it for two reasons: it makes the procedure directly generalizable to nonlinear response surfaces without code modification, and it provides a natural Monte Carlo standard error estimate for $\psi^{(t)}$ via the variance of $\hat{\delta}_m$ across individuals. The inner

loop over $M = 10,000$ individuals is the dominant computational cost, converging at rate $M^{-1/2}$. Monte Carlo standard errors across the $T = 4,000$ outer draws are reported alongside the interval estimates in Section 4.8.

4.6 Sensitivity Analysis

The generative model of Section 4.4 contains three quantities that cannot be identified from the published literature and must be specified as modeling assumptions: the baseline correlation ρ^* between pain and depression in the target population, the baseline means μ^* of the target covariate distribution, and the prior scale σ_β on the slope matrix B . The first two characterize uncertainty about the target population; the third characterizes uncertainty about the source response surface. A systematic sweep over each quantity characterizes how sensitive the transported estimate ψ is to each, and which of the three carries the most weight.

Sweep 1: Baseline correlation ρ^*

Under the linear response surface (4.4), the transported mean ψ depends on the target distribution only through its mean μ^* : equation (4.6) is linear in μ^* , and the covariance structure of $P^*(Z)$, including ρ^* , does not enter. The covariance structure of $P^*(Z)$, including ρ^* , does not enter the expression. Nor does ρ^* affect B : the prior on B is specified independently of the target covariance structure, and the two are drawn separately at each iteration. Varying ρ^* therefore cannot affect ψ analytically; any numerical variation across draws is attributable entirely to Monte Carlo error in the inner loop approximation of μ_{std}^* .

Table 4.1 confirms the analytic prediction. Across $\rho^* \in \{-0.3, 0.0, 0.3, 0.6\}$, the transported pain estimate varies by less than 0.004 points and the depression estimate by less than 0.005 points. The Monte Carlo standard error of the mean estimate is 0.0037 for pain and 0.0032 for depression, identical across all four values of ρ^* , so the observed variation is within one standard error of zero. Both the mean and the variance of $\psi^{(t)}$ are insensitive to ρ^* , consistent with the theoretical result that a linear response surface depends on $P^*(Z)$ only through μ^* .

The stability of ψ across ρ^* is a mathematical consequence of the linearity assumption: under a linear response surface, ψ depends on $P^*(Z)$ only

Table 4.1: Sensitivity of transported outcomes to assumed baseline correlation ρ^* in the target population. Baseline means fixed at reference values $\mu^* = (6.2, 9.1)$.

ρ^*	Pain ψ			Depression ψ		
	Mean	$Q_{0.025}$	$Q_{0.975}$	Mean	$Q_{0.025}$	$Q_{0.975}$
-0.3	-1.273	-1.548	-0.998	-1.853	-2.109	-1.635
0.0	-1.276	-1.550	-1.003	-1.849	-2.106	-1.621
0.3	-1.275	-1.552	-1.007	-1.850	-2.109	-1.625
0.6	-1.274	-1.549	-1.013	-1.851	-2.100	-1.628

through its mean μ^* , so the covariance structure cannot enter. The sweep confirms that the implementation reproduces this theoretical prediction.

Sweep 2: Prior scale σ_β

The prior scale σ_β on slope magnitudes is the second unidentified quantity in the analysis. It governs how much effect heterogeneity by baseline severity the prior on B permits, and because B is not identified from the published moment summaries, σ_β cannot be calibrated against any data quantity. Table 4.2 reports transported estimates across $\sigma_\beta \in \{0.05, 0.10, 0.15, 0.25\}$, a fivefold range centered on the baseline value used in the primary analysis.

Table 4.2: Sensitivity of transported outcomes to the prior scale σ_β on slope magnitudes, with ρ^* held fixed at 0. The center of ψ is stable across the fivefold range while the interval width expands roughly linearly with σ_β .

σ_β	Pain ψ			Depression ψ		
	Mean	$Q_{0.025}$	$Q_{0.975}$	Mean	$Q_{0.025}$	$Q_{0.975}$
0.05	-1.225	-1.319	-1.138	-1.783	-1.870	-1.707
0.10	-1.251	-1.438	-1.067	-1.818	-1.991	-1.669
0.15	-1.277	-1.546	-1.013	-1.851	-2.108	-1.628
0.25	-1.330	-1.797	-0.891	-1.921	-2.336	-1.545

The center of ψ is stable across the fivefold range: the pain mean shifts by

0.105 points and the depression mean by 0.138 points across $\sigma_\beta \in [0.05, 0.25]$, both small relative to the minimally important clinical differences of 1.0 NPRS points and 1.5 HADS points. The interval width, by contrast, expands roughly linearly with σ_β : the 95% pain interval grows from 0.18 points at $\sigma_\beta = 0.05$ to 0.91 points at $\sigma_\beta = 0.25$, and the depression interval grows from 0.16 to 0.79 points. This asymmetry with a stable center and sensitive width is the expected signature of a partially identified model. The anchor constraint fixes α at the published source mean change regardless of which slope matrix is drawn, so the center of ψ does not depend on σ_β . The unidentified shape of B governs how wide the transported interval must be to honestly represent uncertainty about the response surface, so the interval width tracks σ_β directly.

At the lower end of the grid, $\sigma_\beta = 0.05$, the prior is tight enough that the response surface is nearly flat and the transported estimates sit just below the source mean: $\psi_{\text{pain}} = -1.225$ versus $\mu_{\Delta, \text{pain}} = -1.2$, and $\psi_{\text{dep}} = -1.783$ versus $\mu_{\Delta, \text{dep}} = -1.75$. The covariate shift contributes a small correction toward greater improvement even in this conservative regime, but the correction is far smaller than the width of the interval at any σ_β above 0.05.

The depression result is stable across the grid in a clinically meaningful sense: the upper bound of the 95% interval remains at or below -1.545 at every value of σ_β in the grid, so the entire interval exceeds the 1.5 HADS minimally important difference under every prior scale tested, though the margin narrows to within 0.05 points at the most permissive scale. The pain result is more equivocal: at $\sigma_\beta = 0.15$ the upper bound of the interval is -1.013 , essentially at the 1.0 NPRS threshold, and at $\sigma_\beta = 0.25$ the upper bound falls to -0.891 , below the threshold. The robustness of the pain conclusion to the choice of prior scale is therefore more limited than the depression conclusion.

Sweep 3: Target baseline means μ^*

The more consequential sensitivity analysis concerns the target population baseline means. Because μ^* enters ψ directly through equation (4.6), and because it is not identified from the published literature, it represents the primary structural uncertainty in the analysis.

We vary $\mu_{\text{pain}}^* \in \{5.8, 6.2, 6.6\}$ and $\mu_{\text{dep}}^* \in \{8.5, 9.1, 9.7\}$, yielding a 3×3 grid of nine scenarios. The lower bounds represent a target population whose baseline burden is only modestly elevated above the source cohort; the upper

bounds represent a population with substantially higher symptom severity. All scenarios are clinically plausible given the elevated prevalence of chronic pain and depression in TGD adults documented in Tabernacki et al. (2025). The reference scenario (6.2, 9.1) occupies the center of the grid.

Table 4.3: Sensitivity of transported outcomes to assumed target baseline means. Rows vary assumed mean baseline pain and depression in the TGD population across a clinically plausible range. ρ^* fixed at 0.0.

μ_{pain}^*	μ_{dep}^*	Pain ψ			Depression ψ		
		Mean	$Q_{0.025}$	$Q_{0.975}$	Mean	$Q_{0.025}$	$Q_{0.975}$
5.8	8.5	-1.252	-1.452	-1.059	-1.824	-2.009	-1.660
6.2	8.5	-1.278	-1.496	-1.072	-1.823	-2.052	-1.612
6.6	8.5	-1.298	-1.540	-1.076	-1.827	-2.095	-1.562
5.8	9.1	-1.252	-1.513	-0.998	-1.851	-2.076	-1.667
6.2	9.1	-1.276	-1.554	-1.005	-1.847	-2.096	-1.623
6.6	9.1	-1.300	-1.599	-1.016	-1.851	-2.150	-1.577
5.8	9.7	-1.259	-1.592	-0.934	-1.882	-2.152	-1.678
6.2	9.7	-1.277	-1.619	-0.935	-1.882	-2.173	-1.636
6.6	9.7	-1.297	-1.662	-0.964	-1.881	-2.209	-1.600

The pain estimate ψ_{pain} revealed in Table 4.3 is driven almost entirely by μ_{pain}^* : moving from 5.8 to 6.6 shifts ψ_{pain} by approximately 0.05 points regardless of the assumed depression mean. The depression estimate ψ_{dep} is similarly driven almost entirely by μ_{dep}^* : moving from 8.5 to 9.7 shifts ψ_{dep} by approximately 0.06 points regardless of the assumed pain mean.

Cross-outcome effects, the influence of μ_{pain}^* on ψ_{dep} and vice versa, are negligible. The off-diagonal entries of B have mean zero under the prior: $E[B_{12}] = E[B_{21}] = 0$. The contribution of μ_{pain}^* to ψ_{dep} is proportional to $E[B_{21}](\mu_{\text{pain}}^* - \mu_{Z,\text{pain}})$, which cancels in expectation. Cross-outcome slopes contribute only to interval width, not to the center of ψ .

To verify that the decoupling result is not an artifact of fixing $\rho^* = 0$, Table 4.4 replicates the grid at $\rho^* = 0.6$. The pattern is identical: ψ_{pain} responds to μ_{pain}^* but not μ_{dep}^* , and ψ_{dep} responds to μ_{dep}^* but not μ_{pain}^* . Estimates differ by less than 0.01 points across the two tables, within Monte Carlo noise. The decoupling is therefore attributable to the prior on B , specifically, $E[B_{12}] = E[B_{21}] = 0$, rather than to the assumption of zero

baseline correlation.

Table 4.4: Sensitivity of transported outcomes to assumed target baseline means at $\rho^* = 0.6$. Compare with Table 4.3 ($\rho^* = 0.0$): the decoupling pattern is preserved, confirming it is a consequence of $E[B_{12}] = E[B_{21}] = 0$ rather than of the assumed baseline correlation.

μ_{pain}^*	μ_{dep}^*	Pain ψ			Depression ψ		
		Mean	$Q_{0.025}$	$Q_{0.975}$	Mean	$Q_{0.025}$	$Q_{0.975}$
5.8	8.5	-1.255	-1.455	-1.053	-1.823	-2.008	-1.659
6.2	8.5	-1.279	-1.502	-1.070	-1.824	-2.046	-1.615
6.6	8.5	-1.298	-1.542	-1.077	-1.824	-2.096	-1.566
5.8	9.1	-1.254	-1.522	-0.998	-1.850	-2.066	-1.668
6.2	9.1	-1.276	-1.558	-1.002	-1.849	-2.105	-1.626
6.6	9.1	-1.300	-1.597	-1.022	-1.849	-2.146	-1.575
5.8	9.7	-1.260	-1.598	-0.936	-1.882	-2.146	-1.683
6.2	9.7	-1.277	-1.615	-0.938	-1.879	-2.156	-1.648
6.6	9.7	-1.299	-1.659	-0.948	-1.879	-2.197	-1.593

Crucially, the total variation in ψ across all nine scenarios is 0.048 points for pain and 0.059 points for depression. The ranges are an order of magnitude smaller than the width of the 95% prior predictive intervals within any single scenario, which span approximately 0.4–0.7 points. The dominant source of uncertainty in the analysis is therefore uncertainty about the slope matrix B , that is, uncertainty about how baseline severity predicts treatment response in the source population, not uncertainty about the target population baseline means. The limitation that Tabernacki et al. (2025) does not report continuous severity scores has only modest quantitative consequences for the transported estimate.

Joint interpretation

The three sensitivity sweeps cover three distinct sources of uncertainty in the analysis: the joint dependence structure of the target covariate distribution (ρ^*), the marginal location of the target covariate distribution (μ^*), and the prior scale on the source response surface (σ_β). The first two characterize uncertainty about the target population. The third characterizes uncertainty

about the source response mechanism. Together they span the full space of unidentified structural quantities in the model.

The transported estimate is essentially insensitive to assumptions about the target distribution. The ρ^* sweep shows that joint dependence is irrelevant under linearity, as theory predicts. The μ^* sweep shows that the marginal means, while the primary identification gap on the target side, shift ψ by less than 0.05 points for pain and less than 0.06 points for depression across the full nine-scenario grid. By contrast, the σ_β sweep shows that the prior scale on B governs the width of the prior predictive interval almost entirely: as σ_β increases from 0.05 to 0.25, the 95% pain interval expands from 0.18 points to 0.91 points, and the depression interval expands from 0.16 points to 0.79 points. The width of the transported estimate is controlled by the source response surface, not the target covariate distribution.

4.7 The Binding Constraint on Transportability

The sensitivity analysis of Section 4.6 established that the width of the transported interval is governed almost entirely by the prior on B , while the center is essentially insensitive to assumptions about the target population. This pattern has a quantitative consequence that determines what the transported estimate can and cannot say.

At the baseline value $\sigma_\beta = 0.15$, the 95% prior predictive interval for ψ_{pain} spans approximately 0.55 points. The total variation in ψ_{pain} across the entire μ^* grid is approximately 0.048 points, and the transport adjustment from the source mean is roughly 0.08 points. To see where that figure comes from: the covariate shift in source-standardized units is $(\mu_{\text{pain}}^* - \mu_{Z,\text{pain}})/\sigma_{Z,\text{pain}} = (6.2 - 4.8)/2.2 \approx 0.64$ for pain and $(9.1 - 6.9)/2.6 \approx 0.85$ for depression. The prior mean of the diagonal entries of B is $E[B_{ii}] = -\sigma_\beta \sqrt{2/\pi} \approx -0.12$. The expected transport adjustment is therefore $(-0.12)(0.64) \approx -0.077$ for pain and $(-0.12)(0.85) \approx -0.102$ for depression, both near 0.08 to 0.10 in absolute value. The prior standard deviation of the same adjustment term is approximately $\sigma_\beta \times 0.64 \approx 0.096$ for pain, so the adjustment is less than one prior standard deviation from zero. Uncertainty about B is therefore far larger than the transport adjustment itself.

The transported estimate is statistically indistinguishable from the source

mean, and the framework identifies why. The source data do not estimate treatment effect heterogeneity by baseline severity, so the slope matrix B is unidentified, so the transport adjustment cannot be sharper than the prior on B permits. Until a study designed to estimate how baseline severity predicts treatment response exists, the source mean is the defensible point estimate for the TGD population. The framework reaches this conclusion through an explicit identification argument rather than by assuming that the source estimate generalizes.

The depression result is more robust than the pain result. At every value of σ_β in the sweep, the 95% depression interval lies at or below -1.545 , exceeding the 1.5 HADS minimally important clinical difference under every prior scale tested, though the margin narrows to within 0.05 points at the most permissive scale. The transported depression effect is therefore clinically meaningful across the full range of prior assumptions tested. The pain result is more equivocal. At the baseline $\sigma_\beta = 0.15$ the upper bound of the interval is -1.013 , essentially at the 1.0 NPRS threshold, and at $\sigma_\beta = 0.25$ the upper bound falls to -0.891 , below the threshold. The pain conclusion does not survive the prior sensitivity sweep at the same level of robustness.

4.8 Results and Interpretation

Table 4.5 reports the primary transported estimates at the reference scenario $\mu^* = (6.2, 9.1)$ and $\rho^* = 0.0$, summarized as the mean and 95% interval of the constrained prior predictive distribution over ψ .

Table 4.5: Primary transported estimates. Reference scenario: $\mu^* = (6.2, 9.1)$, $\rho^* = 0.0$, $\sigma_\beta = 0.15$, $T = 4,000$, $M = 10,000$. Monte Carlo standard errors in parentheses.

Outcome	$\hat{\psi}$	$Q_{0.025}$	$Q_{0.975}$	MC SE
Pain (NPRS)	-1.275	-1.548	-1.003	(0.012)
Depression (HADS)	-1.850	-2.106	-1.617	(0.016)

Magnitude and direction

Both transported estimates are negative, indicating that ketamine treatment is expected to reduce pain and depression scores in the TGD target population. The transported pain estimate $\hat{\psi}_{\text{pain}} = -1.275$ is modestly larger in magnitude than the source cohort mean change of -1.2 , and similarly, the transported depression estimate $\hat{\psi}_{\text{dep}} = -1.850$ exceeds the source mean of -1.75 . The pattern is theoretically expected: the TGD target population has higher assumed baseline symptom burden than the source cohort, and the clinical monotonicity prior on the diagonal entries of B implies that patients with higher baseline severity experience larger absolute improvements. The covariate shift from source to target therefore acts in the direction of increased treatment response, and the transported estimates reflect it.

The excess over the source mean is small in absolute terms, 0.075 points for pain and 0.100 points for depression, and lies well within the 95% prior predictive interval. The covariate shift is therefore real but modest relative to the uncertainty about the response surface. The transported estimate cannot be sharply distinguished from the source mean effect given current uncertainty about B .

Interval interpretation

The 95% prior predictive interval $(Q_{0.025}, Q_{0.975}) = (-1.548, -1.003)$ for pain and $(-2.106, -1.617)$ for depression quantify uncertainty about the slope matrix B , not sampling uncertainty about a fixed estimand. They should be interpreted as: across the space of response surfaces consistent with the published constraints and the prior on B , 95% yield a transported mean response in that range. The entire 95% interval lies strictly below zero for both outcomes, so under at least 95% of slope matrices drawn from the prior, the transported mean response is a reduction. The direction of the effect is not sensitive to uncertainty about B ; only its magnitude is.

Relation to source cohort estimates

The source cohort reports sex-stratified outcome changes (Pickering et al., 2025). These stratified figures are not used in the transport procedure for two reasons. First, the generative model conditions on continuous baseline severity scores, not on demographic indicators, so sex does not appear as a

covariate in Z . Second, binary sex categories from a cisgender clinical cohort do not map cleanly onto a TGD target population, where gender identity is both the defining characteristic and a source of heterogeneity that a binary variable cannot capture. The transported estimates are therefore anchored to the overall source mean $\mu_{\Delta} = (-1.2, -1.75)$ rather than to any sex-stratified subgroup.

The transported estimates are therefore not direct replications of the source cohort results. Women in the source cohort show mean pain changes of -1.3 and depression changes of -1.6 ; men show -0.8 and -2.2 respectively Pickering et al. (2025). The transported estimates $(-1.275, -1.850)$ are not a weighted average of these sex-stratified figures. They are instead the result of applying the source response surface to a target population with different baseline covariate means, and they reflect the interaction between the covariate shift and the prior on B . The direction of that interaction, larger predicted improvement in the target than in the source, is a consequence of the clinical monotonicity prior, which constrains the diagonal entries of B to be negative. It is not an empirical finding about TGD patients.

Clinical interpretation

The depression result is the clearest finding of the analysis. A transported reduction of 1.850 points on the HADS exceeds the 1.5-point threshold for a minimally important clinical difference (Snaith, 2003), and the 95% prior predictive interval $(Q_{0.025}, Q_{0.975}) = (-2.106, -1.617)$ lies entirely above the threshold in absolute value. This robustness holds not only at the reference scenario but across the full sensitivity analysis: at every value of σ_{β} tested, at every baseline correlation ρ^* , and at every point in the μ^* grid, the 95% interval for depression remains at or below the 1.5-point threshold. No combination of prior and target-population assumptions in the range considered produces a transported depression estimate that fails to clear the clinical threshold. The margin narrows to within 0.05 points at the most permissive prior scale ($\sigma_{\beta} = 0.25$), so the claim is not that the result is unassailable, but that it survives every sensitivity test the framework can construct from the available data.

The pain result is more equivocal. A transported reduction of 1.275 points on the NPRS exceeds the lower boundary of the commonly cited 1.0–2.0-point range for a minimally important clinical difference (Farrar et al., 2001), but the 95% interval $(Q_{0.025}, Q_{0.975}) = (-1.548, -1.003)$ sits almost

exactly at the threshold: the more favorable bound exceeds it by roughly half a point, while the less favorable bound falls essentially on the threshold. Response surfaces on the more favorable end of the prior yield clinically meaningful pain reductions; those on the less favorable end yield reductions at the edge of clinical significance. Under the most permissive prior scale, the upper bound falls below the threshold entirely. The pain conclusion does not survive the sensitivity sweep at the same level of robustness as the depression conclusion.

Chapter 5

Discussion

This thesis asked whether treatment effects observed in a cisgender chronic pain population receiving ketamine could be transported to transgender and gender-diverse adults with chronic pain and depression, a population absent from the clinical trials that generated those effects, conditionally, under specific assumptions. Chapter 3 established identification through S -admissibility and the transportability framework of Bareinboim and Pearl (2016). Chapter 4 computed the transported estimand ψ under constrained prior predictive Monte Carlo and subjected it to systematic sensitivity analysis. The primary finding is that the transported mean response is statistically indistinguishable from the source mean effect μ_Δ : $\hat{\psi}_{\text{pain}} = -1.275$ and $\hat{\psi}_{\text{dep}} = -1.850$, against source means of -1.2 and -1.75 respectively, with 95% prior predictive intervals that span roughly half a point in each outcome. Understanding what that finding means requires understanding why uncertainty about the slope matrix B dominates the analysis by an order of magnitude.

5.1 Explicit Uncertainty as a Scientific Contribution

The near-null result can read as a failure of the framework to produce an answer; but consider the alternatives. In the absence of a formal transportability analysis, the default clinical inference is one of two things: that the source estimate applies to TGD patients without adjustment, or that nothing can be said at all. The first is an implicit assumption of transportability

with no mechanism specified and no conditions checked; it produces the same point estimate as the framework here but offers no account of why that estimate should hold or what would invalidate it. The second forecloses inference entirely, which is its own kind of error when decisions must be made and the source estimate is the only available anchor. A third alternative, a generic sensitivity analysis around the source mean without the transport machinery, would quantify uncertainty but could not say what the uncertainty is uncertainty about: the parameters being varied would have no connection to the identification problem, so the analysis could not say which data would resolve them.

Unfortunately, the framework developed here provides none of these. It produces a transported estimate that is explicit about its identifying assumptions, precise about which structural quantity drives its uncertainty, and specific about what data would resolve that uncertainty. The point estimate may coincide with naive extrapolation, but the epistemic status of the two is not the same.

The S -admissibility analysis in Chapter 3 established the conditions under which the transport formula is valid. Those conditions are falsifiable as they require that baseline severity z renders the selection mechanism ignorable, that the conditional treatment response is structurally invariant across populations, and that the covariate z carries the same meaning across studies. Stating these conditions explicitly converts a vague presumption of generalizability into a set of claims that future data can support or contradict. The sensitivity analysis over σ_β in Chapter 4 identified B as the binding constraint in which the total variation in ψ across the full nine-scenario μ^* grid is an order of magnitude smaller than the width of the 95% prior predictive interval under any single prior scale. Fortunately, this finding is practical and can tell researchers exactly what must be estimated to sharpen the transported result. What is needed is not a more precise characterization of TGD baseline severity, but individual-level data linking baseline severity to treatment response in a source population. An imprecise estimate accompanied by a specific account of what drives the imprecision is more useful than an equally imprecise estimate with no such account.

5.2 Limitations

The limitations that follow demonstrate the framework’s contribution in operation. Each names a structural feature of the available evidence that the framework cannot resolve on its own but can identify precisely enough for future work to address directly.

Four limitations of the analysis are structural rather than incidental, in the sense that they would not be resolved by collecting more data of the same kind.

First, ψ is a transported mean response, not a transported causal contrast. In the absence of a control condition in either study, the analysis cannot separate treatment-attributable improvement from natural symptom trajectory, placebo response, or regression to the mean. The transported estimates characterize expected improvement under the ketamine protocol, not improvement attributable to ketamine specifically.

Second, the structural invariance assumption is untestable from the available data. If TGD adults respond to ketamine differently than cisgender patients at the same baseline severity profile, for reasons not captured by z , the transported estimates will be biased in a direction and magnitude that cannot be determined without outcome data from TGD patients receiving ketamine. Structural invariance is a necessary assumption for transportability in a setting where no such data exist; it cannot be verified.

Third, the identification argument requires that z measures the same construct on the same scale across populations, not merely that both studies include covariates labeled pain and depression.

Fourth, Pickering et al. (2025) is observational rather than randomized, so the no-residual-confounding assumption cannot be guaranteed. If unmeasured prescribing patterns selected sicker patients into ketamine treatment, the source response surface overstates ketamine’s effect, and the transported estimates inherit a bias in the direction of larger apparent improvement. The framework cannot detect this bias from the available data; it can only highlight it.

5.3 Future Directions

The sensitivity analysis and the limitations of Section 5.2 together specify what would sharpen the transported estimate. Each of the three designs

below targets one of the structural constraints the framework identifies: the first relaxes the instrument-equivalence assumption, the second constrains the unidentified slope matrix B , and the third collects target-population outcome data directly and thereby retires the need for transport altogether. The designs are ordered from the most feasible to the most demanding: instrument standardization, constraining B with individual-level data, and direct estimation in the target population.

Instrument alignment

The most straightforward step is to collect continuous severity scores in future TGD cohort studies. A future replication of the Tabernacki et al. design that collected NPRS and HADS scores alongside binary diagnostic prevalence would resolve the instrument-equivalence assumption at a data level. The technical framework for doing so exists: both instruments are freely available, validated across chronic pain and psychiatric populations, and routinely administered in the clinical settings where TGD patients already receive care. Retrospective harmonization using crosswalk tables between established instruments is a partial remedy for existing datasets; prospective standardization is the more reliable solution for future research.

Constraining the slope matrix

B is non-identified from aggregate summary statistics alone. Identifying it requires individual-level data linking baseline severity to treatment response. Two designs are most directly useful.

A small prospective cohort study of TGD adults receiving ketamine treatment at clinical sites with existing infusion programs would provide individual-level pairs (z_i, Δ_i) from within a population overlapping the target. Even a pilot cohort of modest size would constrain B substantially, because the current uncertainty is driven entirely by the absence of any TGD-specific treatment outcome data, not by sample size. Under the simulation framework developed here, such a dataset could be incorporated as an informative prior on B without altering the transport formula.

Alternatively, a ketamine clinic registry with systematic collection of gender identity data alongside continuous NPRS and HADS outcomes at baseline and follow-up would accumulate this information as a byproduct of clinical documentation. Gender identity data are absent from most existing pain

treatment registries; adding them to standard intake forms is a low-cost intervention with high methodological value.

Direct estimation in the target population

The cleanest resolution is a pragmatic clinical trial with targeted enrollment of TGD participants. A Bayesian adaptive design would allow the trial to update estimates of B as TGD-specific data accumulate, beginning with the prior distribution used in this analysis and revising it as outcomes are observed. Such a design formalizes the relationship between the present framework and future data collection: the constrained prior predictive distribution developed here specifies the prior; the adaptive trial specifies the update. When sufficient TGD-specific outcome data exist, transport is no longer necessary, and the framework is retired by its own success.

5.4 Generalizability of the Framework

The transportability pipeline developed here, proceeding from selection diagram construction through S -admissibility analysis, transport formula derivation, and constrained prior predictive simulation, is not specific to ketamine or TGD populations. The same structural problem arises wherever a clinically relevant subpopulation is absent from randomized trials and the available evidence comes from a population that differs on measured covariates.

TGD adults are underrepresented not only in chronic pain research but in antidepressant efficacy trials, cancer screening studies, and cardiovascular risk cohorts. In each of these settings, clinicians and patients must reason about whether findings from the trial population apply to the population in front of them. The framework developed here offers a basis for that reasoning, making the covariate structure of the transport explicit, checking whether identification conditions are satisfied, deriving the form of the adjustment, and quantifying what remains uncertain. Where identification fails, it says so and names the data that would resolve the failure.

The near-null result of this thesis illustrates what that function looks like in practice. The framework produced a specific account of why a sharp transported estimate is not yet available and what data would be needed to obtain one. For a population that has historically been absent from clinical research, that account provides guidance for future data collection.

Appendix A

Supplemental Code

All simulation referenced in Chapter 4 can be found at github.com/zskigen/causal-transport-tgd.

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